

# flashjet

GPU-native jet clustering with the full merge history

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\* speaker

- 1 Jet clustering, and flashjet
- 2 Correctness
- 3 Speed
- 4 Downstream: the clustering tree as tagger input

## Jet clustering, and flashjet

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# Reconstruction is moving to accelerators — clustering has not

## 1. Where the field is

- CMS has run GPUs in the High Level Trigger since Run 3
- **~30 %** of HLT reconstruction is offloaded → **~25 %** less HLT time
- ported: pixel local reco · pixel tracks and vertices · calorimeter local reco

## 2. What is missing

Jet clustering is **still on the CPU**.

## 3. Why it stayed there

- sequential recombination is **IRC safe** — that is why it won
- but each merge **depends on the previous one**
- a serial data dependence is the one thing a GPU does badly

## 4. Why that costs us now

- reclustering **inside a training loop**
- scanning the **jet radius  $R$**
- **substructure** computed on demand

every one of these pays a host ↔ device round trip, once per call

## What it does

- generalised- $k_t$  clustering — anti- $k_t$ ,  $k_t$ , Cambridge–Aachen
- runs on the GPU, **many jets or events per launch**
- returns jets, the **full merge history**, and substructure **as tensors**

## Why it can be done at all

- Cacciari–Salam nearest-neighbour lemma: the smallest  $d_{ij}$  in the *whole event* is always some particle's **geometric** nearest neighbour
  - so each particle only needs to track **its own** NN ...
  - ... and one merge invalidates only  $O(1)$  of them
- ⇒  $O(N^2)$  **instead of**  $O(N^3)$  — the serial chain stays, but each step becomes cheap and parallel

## In the ML ecosystem

- a **PyTorch** call — tensors in, tensors out
- no host ↔ device copy in a training loop

Written in **Triton**: compiles and autotunes for whatever GPU it finds, so there is no separate CUDA build to maintain and results are reproducible once the tuning is cached.

# One exponent, three algorithms

The generalised- $k_t$  distance measure:

$$d_{ij} = \min\left(p_{Ti}^{2p}, p_{Tj}^{2p}\right) \frac{\Delta R_{ij}^2}{R^2}, \quad \Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$$
$$d_{iB} = p_{Ti}^{2p}$$

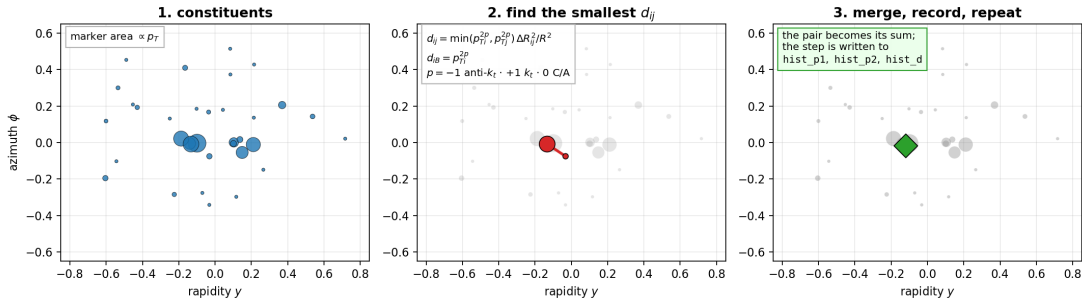
Merge the pair with the smallest  $d_{ij}$ ; if the smallest distance is a  $d_{iB}$ , call  $i$  a jet and remove it.

**Only the exponent  $p$  changes:**

| $p$ | algorithm        | merges first             | tree ordered by       | used for             |
|-----|------------------|--------------------------|-----------------------|----------------------|
| -1  | anti- $k_t$      | the <b>hardest</b> pairs | —                     | <b>finding jets</b>  |
| 0   | Cambridge–Aachen | the <b>closest</b> pair  | <b>angle</b>          | grooming, Lund plane |
| +1  | $k_t$            | the <b>softest</b> pair  | <b>relative</b> $p_T$ | exclusive subjets    |

anti- $k_t$  answers *which particles form a jet*. The other two answer *how that jet came apart* — and the **ordering is what the substructure reads**.

## Sequential recombination — and what flashjet keeps from it



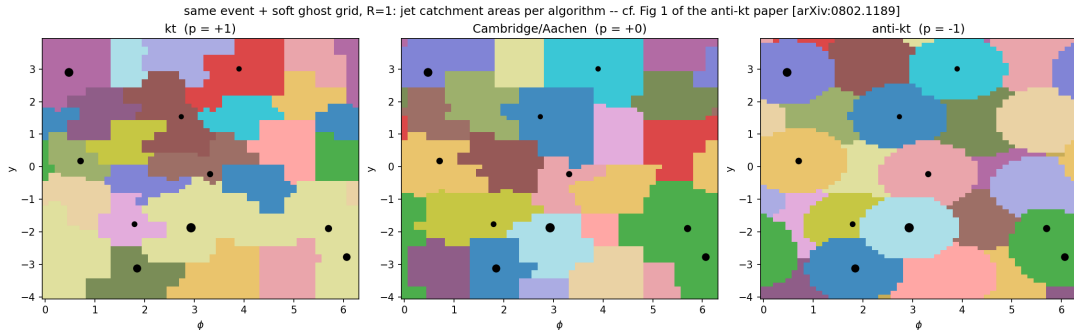
## Two distances, computed for every pair:

- $d_{ij}$  — *pair* distance, particle  $i$  to particle  $j$
- $d_{iB}$  — *beam* distance, particle  $i$  to the beam

## Then repeat until nothing is left:

- ① find the smallest of all distances
- ② if it is a  $d_{ij}$ : **merge**  $i$  and  $j$
- ③ if it is a  $d_{iB}$ : call  $i$  a **jet**, remove it

Both distances were defined on the previous slide; the exponent  $p$  there selects which algorithm this loop runs.



- jet catchment areas, all three algorithms
- clustered by **flashjet** itself

- $k_t$ , C/A: **ragged**, fluctuating boundaries
  - anti- $k_t$ : **circles of area  $\pi R^2$**  on each hard particle
- $\Rightarrow$  the algorithm's defining result, **reproduced without FastJet**

## It returns the tree, not just the jets

Clustering a batch returns **two** things — the jets, and the record of how they were built.

| field            | shape    | what it holds                           |
|------------------|----------|-----------------------------------------|
| jet_idx          | $(B, N)$ | which jet each particle ended up in     |
| n_jets           | $(B,)$   | how many jets were found                |
| hist_p1, hist_p2 | $(B, N)$ | <b>which pair merged</b> , at each step |
| hist_child       | $(B, N)$ | what they merged <i>into</i>            |
| hist_d           | $(B, N)$ | the distance $d_{ij}$ at that step      |

### The tree is the product, not a by-product

The bottom three rows are the **complete binary merge tree**. Groomed mass,  $z_g$ ,  $R_g$ , Lund coordinates and exclusive subjets are not separate algorithms — they are **coordinates on that one tree**. Compute it once, read them all off, **no second clustering pass**.

Three things are read off that tree — F1, F2, F3 — and then we look at it on a real jet.

# What is implemented

## Clustering

- generalised- $k_t$ : anti- $k_t$ ,  $k_t$ , C/A
- merge history recorded in-kernel
- many jets clustered in one batched call
- scales from single jets to whole events

## Substructure — reads of the history

- **F1** exclusive  $k_t$  subjets
- **F2** soft drop / mMDT / mass-drop
- **F3** primary Lund coordinates

All three are pure-torch post-reads: no kernel changes, CPU/GPU identical, negligible cost.

**What it does.** Undo the last merges of the sequence to expose exactly  $n$  subjects — or stop at a  $d_{cut}$ .

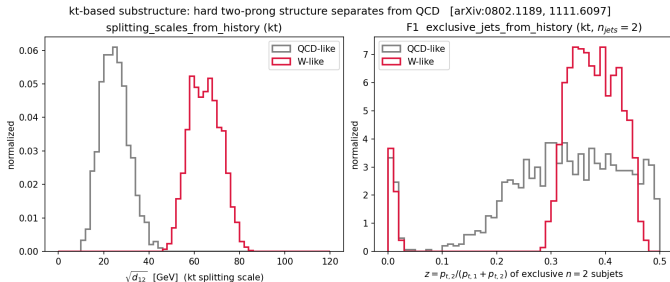
**How.** The history is already a binary tree. Walk back up it  $n - 1$  steps; no reclustering.

**Why  $k_t$ .**  $k_t$  merges the softest pair first, so the *last* merges are the hard prongs — exactly what a 2-prong decay leaves behind.

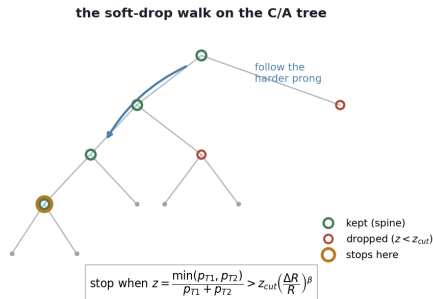
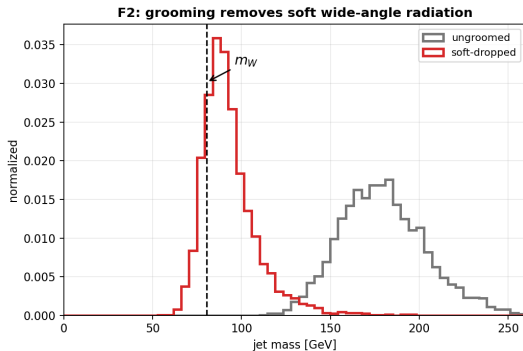
The splitting scales come straight from `hist_d`:

$$\sqrt{d_{12}}, \sqrt{d_{23}}, \dots$$

$$\text{with } d_{ij} = \min(p_{Ti}^2, p_{Tj}^2) \frac{\Delta R_{ij}^2}{R^2}$$



Toy W-like (2-prong) vs QCD-like fat jets. **Left:** the  $k_t$  splitting scale  $\sqrt{d_{12}}$  separates cleanly. **Right:** the momentum balance  $z$  of the two exclusive subjects — the W sits near  $z \sim 0.4$ , QCD spreads to small  $z$ .



**Stop when** 
$$\frac{\min(p_{T1}, p_{T2})}{p_{T1} + p_{T2}} > z_{\text{cut}} \left( \frac{\Delta R_{12}}{R} \right)^\beta$$

Walk down the C/A tree along the harder branch, discarding the softer prong until that condition is met.  $\beta = 0$  recovers mMDT.

The walk is  $O(\log_2 n)$  on the stored tree — it never touches the constituents again, so grooming is essentially free once clustered.

## C/A merge trees of real jets — root at top, constituents at the bottom

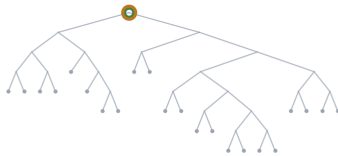
ATLAS Top Tagging Open Data · vertical = declustering depth · horizontal = angular ordering

Light quark / gluon jet



a long spine: soft prong after soft prong  
is stripped, and the mass collapses  
 $n_{\text{drop}} = 8$        $m: 80 \rightarrow 1 \text{ GeV}$

Boosted top — a clean two-prong core



the very first split is already balanced,  
so nothing is groomed away  
 $n_{\text{drop}} = 0$        $m: 80 \rightarrow 80 \text{ GeV}$

Boosted top — the full  $t \rightarrow bW$  system



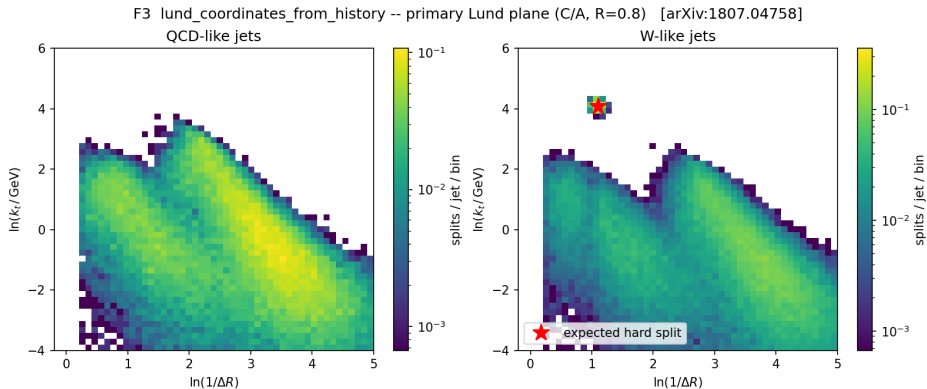
also stops at once, but on a wider split,  
so the whole decay is kept  
 $n_{\text{drop}} = 0$        $m: 170 \rightarrow 170 \text{ GeV}$

● soft-drop spine    ● dropped prong    ● grooming stops here    ● constituent

**A jet without a decay has nothing to stop on.** The light-quark/gluon jet is a long *staircase*: soft prong after soft prong is stripped, and its mass collapses to nothing.

ATLAS Open Data, C/A with soft drop ( $z_{\text{cut}}=0.1$ ,  $\beta=0$ ). **The number of drops is itself a discriminant** — and it is read straight off the stored tree, at no extra cost.

**A real decay stops at once.** Both top jets halt on their very first split, because it is already hard and balanced — so the mass survives.



**What.** Every primary splitting emits  $z = \frac{p_{T2}}{p_{T1} + p_{T2}}$ ,  $\Delta R$ ,  $k_t = p_{T2} \Delta R$ , and the plane coordinates  $(\ln \frac{1}{\Delta R}, \ln k_t)$ .

QCD fills the soft-collinear region smoothly; the W-like sample adds a **localised hard splitting** exactly where the decay is predicted to sit (star).

**How.** One pass down the primary branch — each node is one point in the plane.

**Check.** The  $d$ -channel ties *exactly* to the splitting scales computed independently.

**Correctness**

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# Two kinds of validation

## First: toys, where truth is known

On generated inputs the *answer* is known analytically — so any observable landing off the predicted curve is a real bug, with nothing to hide behind.

**Input A — toy fat jets** — QCD-like and W-like fat jets

**Input B — toy parton shower** — leading-log parton shower

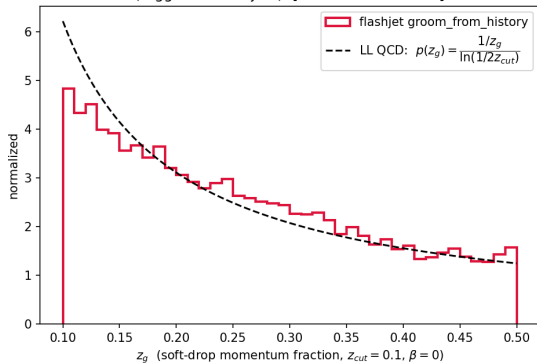
## Then: real data, against FastJet

On detector-level jets there is no analytic answer — so the reference is **FastJet on the same events**, and the experiment's own reconstructed jets.

**Input C — ATLAS Open Data** — large- $R$  jets and full events

The order matters: **toys first** pins the algorithm against a known answer; **real data second** shows it survives detector-level input. A failure at either stage means something different.

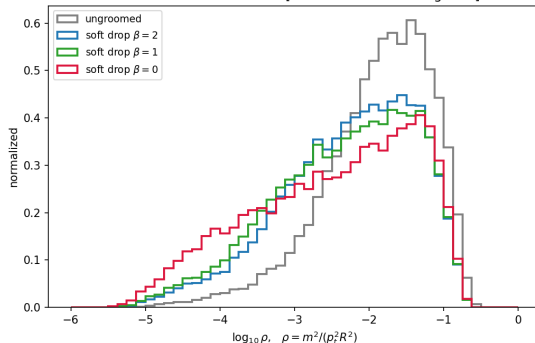
groomed splitting fraction vs the analytic prediction  
(tagged 72% of jets) [cf. arXiv:1402.2657]



Soft-drop  $z_g$  follows the  $1/z$  form predicted at leading-log, recovered from the C/A history.

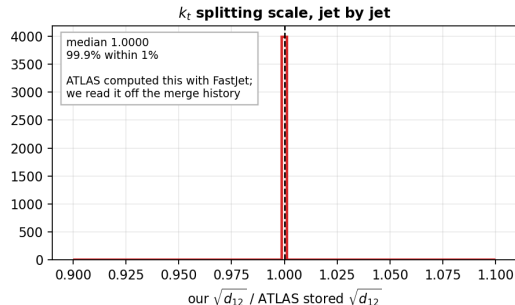
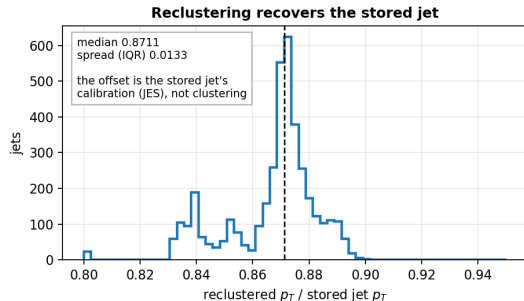
Toy-shower input, so the *prediction* is unambiguous — these test the decoders, not an event generator.

groomed jet mass: stronger grooming (smaller  $\beta$ ) pushes the soft-radiation mass down [cf. arXiv:1402.2657 Figs 3-4]



Groomed-mass ordering in  $\beta$ : larger  $\beta$  grooms less, so the mass peak moves up.

## Jet-by-jet closure against ATLAS-stored quantities



**Left.** Reclustering the stored constituents recovers the stored jet with a **0.013 spread**. The 0.87 offset is the stored jet's **calibration**, applied after clustering — it is present before we touch anything.

**Right.** The  $k_t$  splitting scale  $\sqrt{d_{12}}$  against the value **ATLAS computed with FastJet** and stored: median **1.0000**, **99.9 %** within 1%. We read ours straight off the merge history.

3 algorithms  $\times$  5 radii  $\times$  5 multiplicity bins  $\times$  2 samples, 20 000 jets per bin, on ATLAS Open Data.

| check                              | result                                      |
|------------------------------------|---------------------------------------------|
| number of jets found               | <b>100.000 %</b> at every one of 150 points |
| leading-jet $p_T$ within $10^{-4}$ | 1.0000 at 148/150 points                    |
| median relative difference         | $\sim 7 \times 10^{-8}$ (float32 precision) |

### The grid is the point

A single working point proves little. Agreement holds across **all three algorithms**, **every radius** from 0.4 to 1.2, and **24 to 123** constituents per jet.

### Two independent references

The same events are clustered by **FastJet** for the jets, and every substructure decoder is separately pinned against an **independent NumPy tree-walk**.

A stronger test than agreeing with FastJet.

ATLAS publishes an open dataset carrying both the **calorimeter clusters** and **its own reconstructed jets**. Cluster the  $\sim 600$  clusters per event ourselves, and compare:

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|                            |                      |
|----------------------------|----------------------|
| $n_{\text{jets}}$ vs ATLAS | <b>100.0 %</b> match |
| mean jets/event            | 10.96 vs 10.96       |

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This closes the algorithm against **the experiment's own published output** — not against another implementation of the same algorithm.

599.3 clusters/event: the event regime, where the problem is  $O(N^2)$ .

**Speed**

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**Data** — ATLAS Open Data, freely presentable

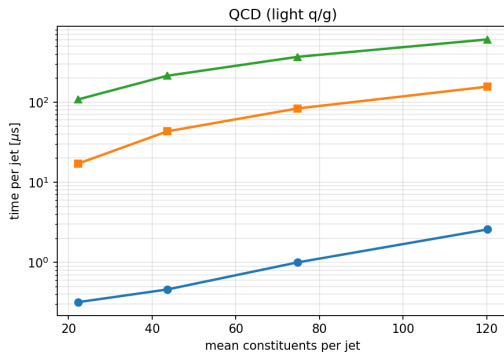
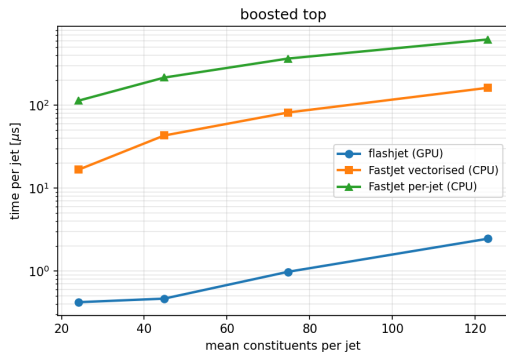
| sample             | jets   | $\langle n_{\text{const}} \rangle$ |
|--------------------|--------|------------------------------------|
| boosted top        | 30 000 | 59.2                               |
| QCD (light $q/g$ ) | 30 000 | 52.4                               |

Large- $R$  jets with calorimeter-cluster constituents. The grid is **3 algorithms**  $\times$  **5 radii**  $\times$  **5 multiplicity bins**, 20 000 jets per bin.

## Method

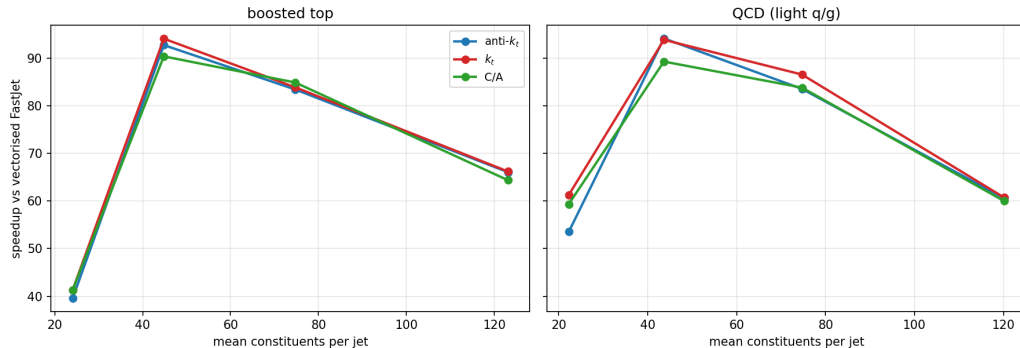
- 1 extract constituents once, save;
- 2 cluster on GPU, time it, save the *same* events;
- 3 cluster those saved events with FastJet;
- 4 **check they agree** — else the timing is meaningless.

Both clusterers read the **same saved file**. Speedups are quoted against the **vectorised** FastJet interface — the faster, fairer baseline.

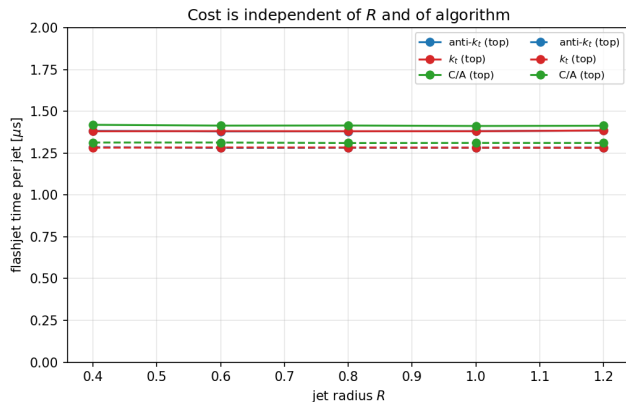
ATLAS Top Tagging Open Data — anti- $k_t$   $R = 1.0$ 

**Log scale** — each gridline is 10×. flashjet stays nearly flat while both FastJet interfaces climb steeply with multiplicity: **0.35–2.7 μs per jet** against **17–155 μs** for vectorised FastJet, and **110–620 μs** per jet one at a time. Benchmarked on an NVIDIA V100-class GPU.

Speedup vs multiplicity —  $R = 1.0$  (all points 100%  $n_{jets}$  agreement)



All three algorithms, both samples,  $R = 1.0$ . **39–99 $\times$**  over vectorised FastJet across the grid (median **64 $\times$** ), and every point has **100 %**  $n_{jets}$  agreement. Busier jets  $\rightarrow$  bigger win — the regime that matters for boosted-object physics.



Across  $R = 0.4 \rightarrow 1.2$ : a **0.3 % spread** over a  $3\times$  range in  $R$ .

$R$  changes *which* pairs merge, not *how many* distances are computed.

Across algorithms: anti- $k_t$ ,  $k_t$  and C/A agree to within  $\sim 4\%$  at every multiplicity.

$k_t$  and C/A cost the same as anti- $k_t$ .

# Using it

```
import flashjet

out = flashjet.cluster(p4, mask, R=1.0, algorithm="antikt")
```

| argument  | type               |                                                   |
|-----------|--------------------|---------------------------------------------------|
| p4        | $(B, N, 4)$ tensor | $(p_x, p_y, p_z, E)$ ; CPU or CUDA                |
| mask      | $(B, N)$ bool      | marks real constituents in a padded batch         |
| R         | float              | jet radius                                        |
| algorithm | str                | "antikt"   "kt"   "cambridge"                     |
| p         | float              | generalised- $k_t$ exponent (overrides algorithm) |

|                                              |           |                                         |
|----------------------------------------------|-----------|-----------------------------------------|
| out.n_jets                                   | # (B,)    | jets found                              |
| out.jets_p4(p4)                              | # (B,J,4) | jet four-momenta                        |
| out.splitting_scales()                       | #         | $\sqrt{d_{12}}$ , $\sqrt{d_{23}}$ , ... |
| out.exclusive_jets(n_jets=3)                 | # F1      | exclusive $k_t$ subjects                |
| out.groomed_jets(p4, R, z_cut=0.1, beta=0.0) | # F2      | soft drop / mMDT                        |
| out.lund_coordinates(p4, R)                  | # F3      | primary Lund plane                      |

The same call runs unchanged on CPU and GPU — the backend is chosen from the problem size, not by the caller.

**Downstream: the clustering tree as tagger input**

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# Does the tree help a tagger? — setup

**The question.** Gouskos & Maier [PLuM, 2605.26821] report that explicit Lund-plane splittings help a transformer tagger. We test that on a **full 10-class** training.

**Model.** Particle Transformer [2202.03772] — 8 encoder + 2 class-attention blocks, 8 heads, embed 128. **2.144 M** params baseline, **2.194 M** with the feature block (+0.051 M).

**Data.** JetClass, 10 classes, **~20.05 M** test jets ( $\sim 2.0$  M/class), 128 particles/jet. All arms read **identical** shards; features computed live.

Ranger, lr  $10^{-3}$ , batch 512,  $10^6$  iterations, checkpoint every 20k. Uncertainties: bootstrap of the test set, 16–84th percentile.

## The feature arms

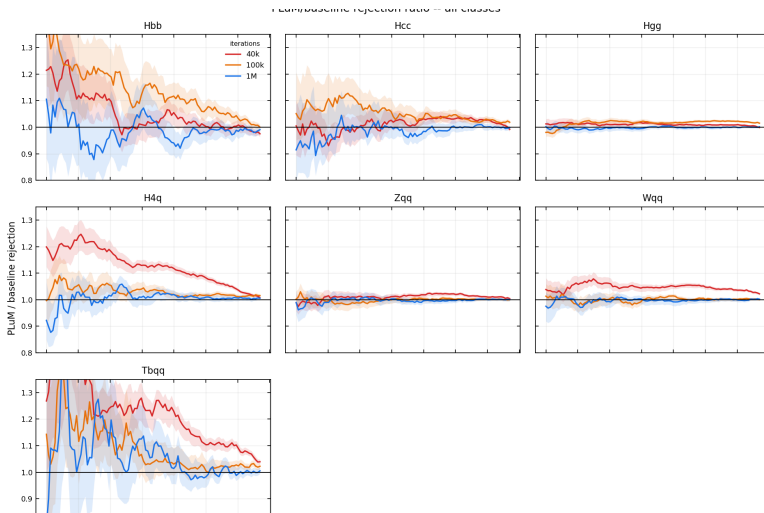
- **PLuM** — 48  $k_t$ -ordered Lund splittings,  $[\ln k_t, \ln 1/\Delta R, \ln z]$ , embedded and **concatenated** as extra tokens (176 vs 128)
- **subject** —  $n$ -body subject mass,  $p_T$  fraction,  $n_{\text{const}}$
- **CA5** — per-particle C/A branch-point features

## Cost (wall clock per step)

|         |       |
|---------|-------|
| PLuM    | 1.59× |
| subject | 1.42× |
| CA5     | 1.22× |

One seed per arm — see the caveat on the next slide.

# Does the tree help a tagger? — result



PLuM/baseline rejection ratio; unpaired-bootstrap bands.

Test accuracy  $\sim 20.05$  M jets

|          | 40k    | 100k   | 1M            |
|----------|--------|--------|---------------|
| baseline | 83.431 | 84.632 | <b>86.212</b> |
| PLuM     | 83.428 | 84.697 | 86.206        |
| $\Delta$ | -0.003 | +0.065 | -0.006        |

## What we see

An early gain in every class, gone by 1 M. Wqq has the *fewest* splittings (21.6/jet) yet gains at  $\sim 9\sigma$ : **not confined to complex or *b*-jets**. At convergence: **null-or-worse**.

## What we do *not* claim

- **No mechanism** — *b*-specificity excluded.
- **One seed per arm** — a second seed's spread exceeded the effect.
- Their Hbb reproduced at 100k, **not** at convergence.

- 1 **Clusters on the GPU and keeps the full merge history**, so substructure — exclusive subjects, soft drop, Lund coordinates — comes free as array reads.
- 2 **Gives the same jets as FastJet.** 150/150 grid points at 100 %  $n_{\text{jets}}$  agreement, across anti- $k_t/k_t$ /C-A,  $R = 0.4\text{--}1.2$  and 24–123 constituents per jet.
- 3 **Reproduces the experiment's own reconstructed jets** exactly, at  $\sim 600$  clusters per event.
- 4 **39–99 $\times$  faster** than vectorised FastJet, and the advantage *grows* with jet complexity. Cost is independent of  $R$  and of algorithm.
- 5 Still a **prototype**, with meaningful headroom left.

**Thank you**

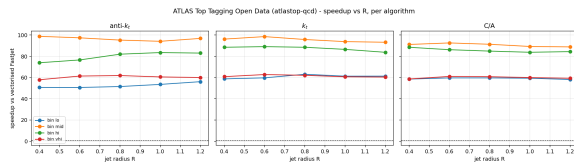
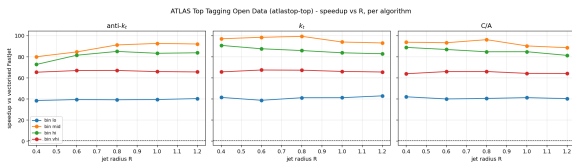
Questions?

Benchmarks: ATLAS Open Data (CC0).

Physics closures: analytic predictions on toy showers.

## Backup

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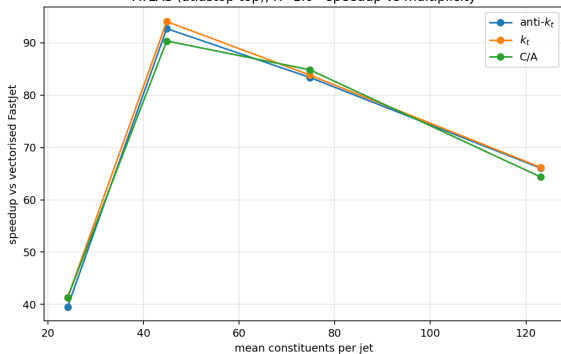


boosted top

Flat in  $R$  for all three algorithms; ordering is by multiplicity bin.

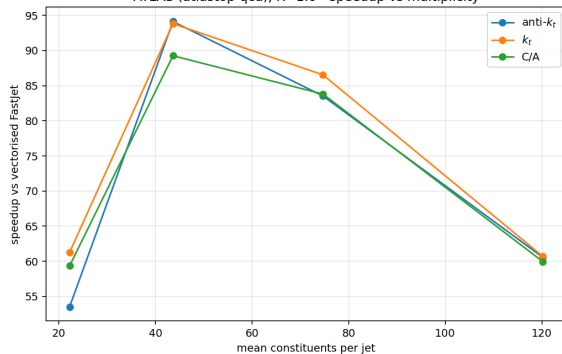
QCD (light  $q/g$ )

ATLAS (atlastop-top), R=1.0 - speedup vs multiplicity



boosted top

ATLAS (atlastop-qcd), R=1.0 - speedup vs multiplicity



QCD (light  $q/g$ )

## Backup — the numbers

| sample | alg         | $\langle n \rangle$ | flashjet<br>[ $\mu\text{s}/\text{jet}$ ] | FastJet vec.<br>[ $\mu\text{s}/\text{jet}$ ] | speedup     |
|--------|-------------|---------------------|------------------------------------------|----------------------------------------------|-------------|
| top    | anti- $k_t$ | 24.1                | 0.427                                    | 16.5                                         | 39×         |
| top    | anti- $k_t$ | 44.8                | 0.464                                    | 43.0                                         | <b>93</b> × |
| top    | C/A         | 74.7                | 1.000                                    | 84.8                                         | 85×         |
| top    | C/A         | 123.1               | 2.565                                    | 165.1                                        | 64×         |
| qcd    | anti- $k_t$ | 22.2                | 0.346                                    | 17.5                                         | 51×         |
| qcd    | anti- $k_t$ | 43.6                | 0.463                                    | 45.7                                         | <b>99</b> × |
| qcd    | anti- $k_t$ | 74.7                | 1.215                                    | 89.8                                         | 74×         |
| qcd    | anti- $k_t$ | 120.2               | 2.673                                    | 154.9                                        | 58×         |

**39–99×** over vectorised FastJet (median 64×), with 100 % jet-count agreement everywhere. Benchmarked on an **NVIDIA V100-class** GPU.

$R = 1.0$  rows shown. Against the per-jet FastJet interface the gap is roughly *two orders of magnitude*.

## Backup — where the time goes

### Nsight Systems — share of GPU time

| kernel                | % GPU       | avg/call   |
|-----------------------|-------------|------------|
| _cluster_large_kernel | <b>97.6</b> | 5.47 ms    |
| _decode_kernel        | 1.5         | 82 $\mu$ s |
| torch glue            | < 1         | —          |

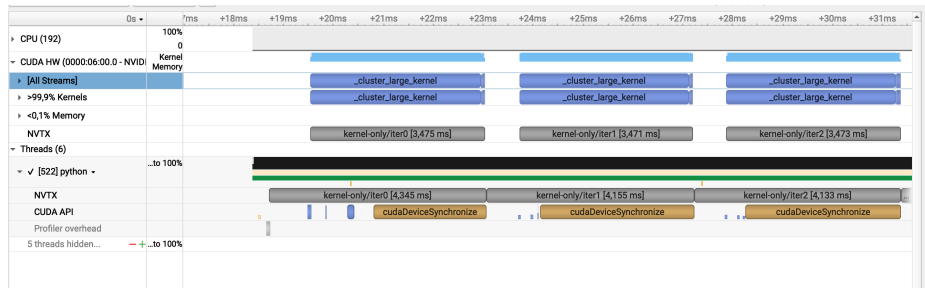
**No hidden overhead** — essentially all the time is real clustering work, and the memory row of the timeline is empty.

### Nsight Compute — hardware counters

| metric                  | value         |
|-------------------------|---------------|
| DRAM throughput         | <b>0.02 %</b> |
| Compute (SM) throughput | 15.4 %        |
| Registers / thread      | 168           |
| Theoretical occupancy   | 18.75 %       |
| Achieved occupancy      | <b>6.25 %</b> |
| Waves per SM            | <b>0.32</b>   |

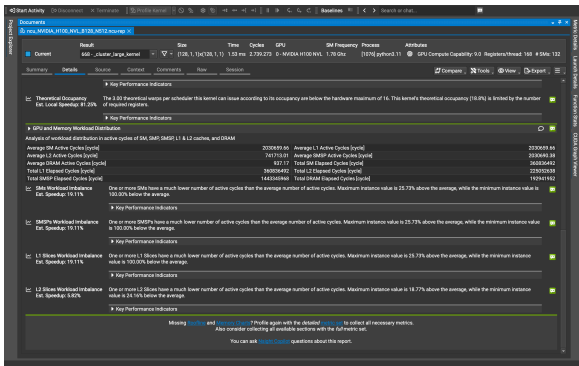
Not memory- or compute-bound — the GPU is simply **not filled**.

# Backup — the GPU timeline



Nsight Systems, three consecutive iterations. Each block is one clustering call; the tool labels the rows “>99.9 % Kernels” and “<0.1 % Memory”. The kernels run back to back with nothing in between and the memory row is empty — no host/device shuffling while clustering runs. That is what makes the speedup real rather than an artefact of where the timer was started.

# Backup — the profiler names the bottleneck



## The tool's own speedup estimates:

finding est. speedup

Theoretical occupancy (registers) 81 %

Achieved occupancy 67 %

SM workload imbalance 19 %

*"This kernel grid is too small to fill the available resources on this device, resulting in only 0.32 full waves across all SMs."*

Grid size **128 blocks** on a **132-SM** device — fewer blocks than the GPU has processors.

~**1.7–1.8×** more available from launch configuration and register budget — **not** from the algorithm.

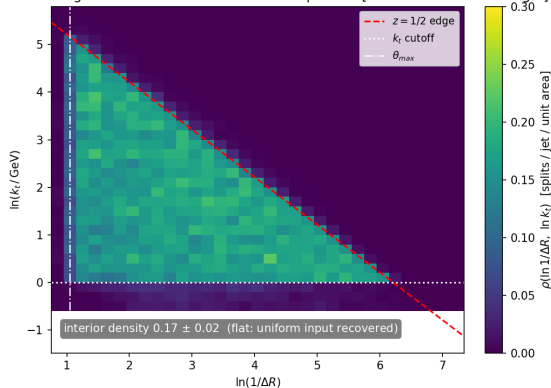
# Backup — occupancy detail

| Occupancy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------|
| Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicate highly imbalanced workloads. |       |                                |
| Theoretical Occupancy (%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 18.75 | Block Limit Registers [block]  |
| Theoretical Active Warps per SM [warp]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 16    | Block Limit Shared Mem [block] |
| Achieved Occupancy (%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 6.25  | Block Limit Warps [block]      |
| Achieved Active Warps per SM [warp]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.00  | Block Limit SM [block]         |
| Cluster Occupancy (%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0     | Block Limit Registers [block]  |
| Max Active Clusters [cluster]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0     | Max Cluster Size [block]       |
| Overall GPU Occupancy (%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0     |                                |
| The difference between calculated theoretical (18.75%) and measured achieved occupancy (6.25%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the <a href="#">Occupancy Guide</a> for more details on optimizing occupancy.                                                                                                                                                                               |       |                                |
| Key Performance Indicators                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |                                |
| The 3.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 16. This kernel's theoretical occupancy (18.8%) is limited by the number of required registers.                                                                                                                                                                                                                                                                                                                                           |       |                                |
| Key Performance Indicators                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |                                |

| Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------|
| Warning: Data collection happened without GPU frequencies being set by the profiler. Some results may be inaccurate.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |                                              |
| GPU Speed (0 Light Throughput)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |                                              |
| High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-items of Compute and Memory to clearly identify the highest constraint.                                                                                                                                                                                                                               |           |                                              |
| Compute (SM) Throughput [N]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 15.40     | Duration [ns]                                |
| Memory Throughput [N]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 17.25     | Elapsed Cycles [cycle]                       |
| L1TEX Cache Throughput [N]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 33.23     | SM Active Cycles [cycle]                     |
| L2 Cache Throughput [N]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.82      | SM Frequency [GHz]                           |
| DRAM Throughput [N]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0.02      | DRAM Frequency [Hz]                          |
| Small Grid: This kernel grid is too small to fill the available resources on this device, resulting in only 6.32 full warps across all SMs. Look at the <a href="#">Launch Statistics</a> for more details.                                                                                                                                                                                                                                                                                                                                                           |           |                                              |
| Key Performance Indicators                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |                                              |
| Launch Statistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |                                              |
| Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.                                                                                                                                                                                                                                                                              |           |                                              |
| Grid Size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 128       | Function Cache Configuration                 |
| Cluster Size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0         | Registers Per Thread [register/thread]       |
| Cluster Scheduling Policy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Polygraph | Static Shared Memory Per Block [byte/block]  |
| Block Size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 128       | Dynamic Shared Memory Per Block [byte/block] |
| Threads [thread]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 16384     | Driver Shared Memory Per Block [byte/block]  |
| Waves per SM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.32      | Shared Memory Configuration Size [byte]      |
| Warp Queue Context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0         | Warp Size                                    |
| # SMs (SM)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 132       | #TPCs                                        |
| Enabled TPCs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | all       |                                              |
| Small Grid: The grid for this launch is configured to execute only 128 blocks, which is less than the 132 multiprocessors used. This can underutilize some multiprocessors. If you do not intend to execute this kernel concurrently with other workloads, consider reducing the block size to have at least one block per multiprocessor or increase the size of the grid to fully utilize the available hardware resources. See the <a href="#">Workload Model</a> description for more details on launch configurations.                                           |           |                                              |
| Key Performance Indicators                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |                                              |
| Occupancy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |                                              |
| Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicate highly imbalanced workloads. |           |                                              |
| Theoretical Occupancy (%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 18.75     | Block Limit Registers [block]                |

Register pressure (168/thread) caps theoretical occupancy at 18.75 %; achieved is 6.25 %. With 0.32 waves/SM the kernel does not fill the GPU even once, while DRAM throughput is 0.02 % — this is parallelism starvation, not a bandwidth problem. Both are configuration inherited from smaller GPUs, not properties of the algorithm.

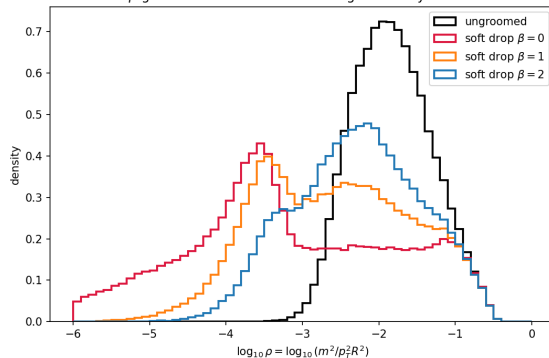
primary Lund plane of the toy shower (C/A, R=0.4):  
m-in-triangle emissions come back as a flat plateau [cf. arXiv:1807.04758 Fig 2]



The Lund plane is populated **uniformly** inside the kinematic **triangle** — the leading-log expectation.

Independent NumPy tree-walks pin every decoder, and the  $d$ -channel of the Lund output ties *exactly* to the splitting scales computed separately.

Soft-drop  $\beta$ -ordering on REAL QCD jets (164292 jets,  $z_{cut} = 0.1$ )  
smaller  $\beta$  grooms harder — same ordering as the toy-shower closure



The soft-drop  $\beta$ -family orders as predicted: larger  $\beta$  grooms less.

## Backup — Input A: the toy fat jets

**Purpose.** QCD vs W is a *known* truth, so any observable that fails to separate them — or lands off the analytic curve — is a real bug.

Two single-fat-jet samples at  $p_T \approx 500$  GeV,  $R = 0.8$ :

- **QCD-like:** one hard collinear core (92 % of  $p_T$ ,  $\sigma_{y,\phi} = 0.06$ ) + soft wide-angle radiation (8 %,  $\sigma = 0.30$ )
- **W-like:** two hard prongs with pair mass  $m = \sqrt{z(1-z)} p_T$ ,  $\Delta R = 80.4$  GeV,  $z \in [0.30, 0.45]$ , random orientation, + 6 % soft contamination

Each “spray” is  $n$  massive pions with exponential  $p_T$  fractions:

```
frac = rng.exponential(1., n); frac /= frac.sum()
pt    = pt_total * frac
y, phi = rng.normal(y0, sigma, n), rng.normal(phi0, sigma, n)
mt     = np.sqrt(M_PI**2 + pt**2)
```

1500 events per sample. `flashjet` has **no event generator** — it clusters four-vectors; these toys live in the plot scripts, outside the library.

## Backup — Input B: the toy parton shower

A **primary, fixed-coupling, leading-log** shower: emissions sampled **uniformly in the Lund triangle** with density  $\bar{\alpha}$  per unit  $(\ln 1/\theta, \ln k_t)$  area, off a hard spine.

$$\theta = e^{-u}, \quad k_t = e^v, \quad z = \frac{k_t}{p_{T0} \theta}$$

keep  $v \leq \ln(p_{T0}/2) - u$  (i.e.  $z \leq \frac{1}{2}$ ),  $k_t > k_t^{\min}$

### Why this is the right test

The generator's density is *uniform in the triangle by construction*, so the leading-log predictions — the  $1/z$  form of  $z_g$ , the  $\beta$ -ordering of the groomed mass — are unambiguous. Landing on them tests the **decoders**, not the shower.

**Jet areas** use the anti- $k_t$ -paper construction: one hard event plus a dense grid of **infinitely soft ghosts** ( $p_T = 10^{-8}$ ); the ghosts each algorithm sweeps up trace its catchment area.

$\bar{\alpha} = 0.25$ ,  $p_{T0} = 1$  TeV,  $R = 0.4$ , 20 000 showers; fixed seed, fully re-runnable.

# Backup — the validation ladder

| rung                    | input  | reference                                | result                                  |
|-------------------------|--------|------------------------------------------|-----------------------------------------|
| unit tests              | —      | independent NumPy tree-walks             | 129 pass, incl. GPU paths               |
| anti- $k_t$ jet areas   | toy B  | the anti- $k_t$ paper's own construction | rigid cones, $\rightarrow \pi R^2$      |
| $k_t$ substructure      | toy A  | known 2-prong vs QCD truth               | clean separation                        |
| soft-drop $z_g$         | toy B  | leading-log $1/z$ prediction             | on the curve                            |
| groomed mass vs $\beta$ | toy B  | Soft Drop Figs. 3–4                      | ordering reproduced                     |
| Lund plane              | toy A  | primary-Lund construction                | hard split where predicted              |
| jet-by-jet kinematics   | real C | FastJet, same events                     | 150/150 points, 100 % $n_{\text{jets}}$ |
| full-event clustering   | real C | the experiment's own jets                | 100 % match, $\sim 600$ clusters/ev     |

**Each rung is closed against something independent** — an analytic prediction, a separate NumPy implementation, or FastJet itself. No rung is validated against another part of flashjet.

- H. Qu, C. Li, S. Qian, *Particle Transformer for Jet Tagging*, arXiv:**2202.03772**. (also the JetClass dataset; Zenodo 10.5281/zenodo.6619768)
- L. Gouskos, B. Maier, *Particle-Lund Multimodality in Jet Taggers*, arXiv:**2605.26821**.
- H. Qu, L. Gouskos, *Jet tagging via particle clouds* (ParticleNet), Phys. Rev. D **101** (2020) 056019, arXiv:1902.08570.
- F. A. Dreyer, G. P. Salam, G. Soyez, *The Lund Jet Plane*, JHEP **12** (2018) 064, arXiv:1807.04758.
- F. A. Dreyer, H. Qu, *Jet tagging in the Lund plane with graph networks*, JHEP **03** (2021) 052, arXiv:2012.08526.
- A. J. Larkoski, S. Marzani, G. Soyez, J. Thaler, *Soft Drop*, JHEP **05** (2014) 146, arXiv:1402.2657.
- M. Cacciari, G. P. Salam, G. Soyez, *The anti- $k_t$  jet clustering algorithm*, JHEP **04** (2008) 063, arXiv:0802.1189; *FastJet user manual*, Eur. Phys. J. C **72** (2012) 1896, arXiv:1111.6097.
- S. Catani, Yu. L. Dokshitzer, M. H. Seymour, B. R. Webber, Nucl. Phys. B **406** (1993) 187 ( $k_t$ ); Yu. L. Dokshitzer *et al.*, hep-ph/9707323 and M. Wobisch, T. Wengler, hep-ph/9907280 (Cambridge/Aachen).
- J. de Favereau *et al.*, *DELPHES 3*, JHEP **02** (2014) 057, arXiv:1307.6346.

All benchmark data is **ATLAS Open Data** (CC0), read over the public redirector — no grid certificate required.

| dataset                       | used for                             |
|-------------------------------|--------------------------------------|
| ATLAS Top Tagging Open Data   | timing + FastJet agreement           |
| 2020 ATLAS Jet Reconstruction | closure vs the experiment's own jets |

Constituents are calorimeter clusters, massless. Physics closures use toy showers compared against leading-log analytic predictions, so the prediction being tested is unambiguous.